K L (DEEMED TO BE) UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

HEALTHSCOPE – A FULL-LIFECYCLE PUBLIC HEALTH ANALYTICS & RISK DASHBOARD

A PROJECT REPORT

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in Computer Science and Engineering

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Declaration

The Project Report entitled "**HEALTHSCOPE – A FULL-LIFECYCLE PUBLIC HEALTH ANALYTICS & RISK DASHBOARD**" is a record of Bonafide work of 2420030697 (D.Poojitha), 2420030322 (P. Kavya Sai), 2420030400 (Masuma Fathema) and 2420030588 (Ch .Bharghavi) submitted in partial fulfillment for the award of B. Tech in CSE to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is certify that the mini project based report entitled "**HEALTHSCOPE – A FULL-LIFECYCLE PUBLIC HEALTH ANALYTICS & RISK DASHBOARD "** is a bonafide work done and submitted by 2420030697 (D.Poojitha), 2420030322 (P. Kavya Sai), 2420030400 (Masuma Fathema) and 2420030588 (Ch .Bharghavi) in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of CSE, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Guide

Signature of the Course Coordinator **Signature of the HOD**

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ABSTRACT

HealthScope is a full-lifecycle public-health analytics and risk-assessment project that combines healthcare data preparation, exploratory data analysis, machine-learning classification and an interactive Streamlit interface. The system classifies the presence or absence of three health conditions – diabetes, heart disease and hypertension – using three publicly available datasets: the Pima Indians Diabetes Database (768 records), the UCI Heart Disease Cleveland subset (303 records) and the Framingham Heart Study dataset (4,240 records).

The workflow is implemented in Python with Pandas, NumPy and scikit-learn. Missing values are handled with SimpleImputer (median strategy), zero values that are physiologically implausible in the diabetes dataset are treated as missing, and numerical features are standardised with StandardScaler. Each dataset is split 80:20 with stratification and random_state = 42. Logistic Regression (max_iter =

compared using accuracy, precision, recall, F1-score and confusion matrices. The final model for each condition is chosen with a weighted score of 70% F1-score and 30% accuracy.

On the held-out test sets, the Decision Tree achieved the best results: 75.97% accuracy and 67.83% F1-score for diabetes (154 test records), 81.97% accuracy and 85.71% F1-score for heart disease (61 test records), and 89.39% accuracy and 83.64% F1-score for hypertension (848 test records). The Streamlit interface provides a home page, a disease selector, a health-input form, a risk estimate, comparisons with reference ranges, visual insights and suggested next steps. The 51.7% value displayed in the example dashboard screenshot is an individual risk estimate and not a model-accuracy score.

The models estimate dataset-defined disease status rather than future onset, they have not been validated on independent populations, and the hypertension model uses blood-pressure measurements as inputs, which limits how its score should be interpreted. HealthScope is therefore an educational prototype and not a diagnostic medical device.

Keywords: Public health analytics; diabetes; heart disease; hypertension; machine learning; Decision Tree; Random Forest; Streamlit; risk assessment.

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CHAPTER 1 – INTRODUCTION

1.1 Background

Chronic non-communicable conditions such as diabetes, heart disease and hypertension are among the most significant public-health challenges of the present day. They develop gradually, are influenced by many interacting factors – age, blood glucose, blood pressure, body-mass index (BMI), cholesterol, smoking and family history – and are often detected late, after complications have begun. Because the same indicators contribute to more than one condition, a person at risk for one disease frequently carries risk for another.

Healthcare datasets collected by clinical studies and surveys contain valuable population-level signals about these relationships. However, the patterns are difficult to identify through manual inspection of rows and columns. Data science provides a structured process – data cleaning, exploratory analysis, statistical summarisation and predictive modelling – for turning such records into insight. Machine-learning classifiers in particular can learn the combinations of measurements that distinguish individuals with a condition from those without it in a given dataset.

HealthScope applies this process to three conditions in a single project. It covers the whole analytics lifecycle: acquiring public datasets, preparing and exploring the data, training and comparing classification models, selecting a model for each condition, and presenting the outcome through an interactive web dashboard that a non-technical user can operate.

1.2 Problem Statement

Healthcare data are abundant but hard to interpret, and the tools that analyse them are usually separate from the tools that present results to users. Analysts work in notebooks, while the people who might benefit from an educational view of their own health indicators have no accessible interface. In addition, projects often study one disease at a time, which makes it harder to compare modelling approaches consistently.

The problem addressed by this project is to design and implement an end-to-end system that (a) prepares and explores three health datasets, (b) trains and objectively compares several classification algorithms for diabetes, heart disease and hypertension, and (c) exposes the selected models through an easy-to-use dashboard that accepts health measurements and returns an educational assessment with visual context.

1.3 Motivation

- **Early awareness:** many people are unaware of how their own glucose, blood pressure or BMI values compare with reference ranges. An accessible tool can raise awareness and encourage timely consultation with a healthcare professional.
- **Consistent comparison:** applying the same preprocessing, split and evaluation protocol to three datasets makes model behaviour across conditions comparable.
- **Full-lifecycle learning:** the project demonstrates how data engineering, machine learning and user-interface development fit together in one deliverable, which is the central idea behind the “full-lifecycle” in the project title.
- **Transparent, interpretable models:** the use of Decision Tree, Logistic Regression and Random Forest keeps the modelling approach explainable and appropriate for an academic demonstration.

1.4 Scope

HealthScope addresses **multiple health outcomes** rather than cardiovascular risk alone. Its classification models estimate **dataset-defined disease status** – the presence or absence of diabetes, heart disease or hypertension as labelled in each source dataset. They do not establish future disease onset, provide clinically validated individual risk probabilities, or replace professional medical diagnosis.

The scope of the work covers dataset integration, preprocessing, exploratory data analysis, model training and evaluation, model selection and a Streamlit dashboard. External validation with independent datasets, clinical validation and production deployment are outside the scope of the work completed so far and are recorded as future work.

1.5 Objectives

1. **Diabetes prediction:** classify whether an individual is likely to have diabetes from features such as glucose level, BMI, age, blood pressure and other health indicators.
2. **Heart disease prediction:** classify the presence or absence of heart disease from clinical features such as age, cholesterol, resting blood pressure, maximum heart rate and other cardiovascular indicators.
3. **Hypertension prediction:** classify the presence or absence of hypertension from features such as age, systolic and diastolic blood pressure, BMI, glucose and other health indicators.
4. **Health data analysis:** perform exploratory data analysis (EDA) to understand feature distributions, identify patterns and examine relationships between health indicators and target outcomes.
5. **Model evaluation and comparison:** compare Logistic Regression, Decision Tree and Random Forest classifiers using accuracy, precision, recall and F1-score.
6. **Interactive health assessment:** integrate the prediction models and visualisations into a Streamlit dashboard that lets users enter health information and view educational assessment results.

1.6 Importance and Target Users

The prototype is intended for academic demonstration, for data-science learners, and for users seeking an accessible educational view of selected health indicators. It can also serve as a starting template for larger public-health analytics dashboards. It is not intended to diagnose disease, prescribe treatment or replace a qualified healthcare professional, and the interface states this explicitly.

1.7 Organisation of the Report

Table 1.1 – Organisation of the report

Chapter	Focus
1	Introduction – background, problem statement, motivation, scope and objectives
2	Literature Survey – related work on machine-learning prediction of diabetes and cardiovascular conditions
3	System and Data Requirements – hardware, software and dataset inventory
4	Technology Stack – libraries and tools used and their roles
5	System Architecture – end-to-end data and model flow
6	Data Understanding and Preparation – datasets, features, targets and preprocessing
7	Exploratory Analysis and Feature Engineering – distributions, correlations and feature importance
8	Machine Learning Methodology – algorithms, hyperparameters, evaluation and model selection
9	Implementation and User Interface – dashboard pages and assessment flow
10	Results and Performance Analysis – metrics, confusion matrices and comparison
11	Testing and Validation – tests performed and planned
12	Challenges and Limitations
13	Conclusion and Future Scope
14	References and Appendices

CHAPTER 2 – LITERATURE SURVEY

2.1 Overview

Machine-learning prediction of chronic disease has been studied extensively. The studies surveyed for this project fall into two groups: work on diabetes classification and risk stratification, and work on explainable machine learning for cardiovascular risk. Together they motivated three design decisions in HealthScope: comparing several classical classifiers instead of relying on one, paying explicit attention to missing values, and favouring models whose behaviour can be explained.

2.2 Diabetes Classification and Risk Prediction

Maniruzzaman et al. [1] investigated the classification and prediction of diabetes using a machine-learning paradigm, comparing several classifiers on diabetes data. In a related study [2], the same group examined accurate diabetes risk stratification with attention to the role of **missing values and outliers**, which is directly relevant to datasets such as the Pima Indians Diabetes Database where several clinical measurements are recorded as zero when they were not actually measured. HealthScope addresses the same issue by treating zero values in Glucose, BloodPressure, SkinThickness, Insulin and BMI as missing and imputing them.

Xie et al. [3] built risk prediction models for type 2 diabetes using machine-learning techniques, illustrating that routinely collected health variables can support risk models. Thotad et al. [4] applied machine-learning methods to detect and classify diabetes on Indian demographic and health survey data, showing the use of survey-based data rather than clinical measurements alone. Joshi and Dhakal [7] predicted type 2 diabetes using logistic regression and other machine-learning approaches, supporting the use of logistic regression as an interpretable baseline – the role it plays in HealthScope.

Lai et al. [8] developed predictive models for diabetes mellitus using machine-learning techniques, and Chang et al. [9] focused on the Pima Indians dataset, classifying diabetes mellitus with several machine-learning algorithms – the same dataset used for the diabetes module of this project. Tasin et al. [10] combined diabetes prediction with explainable-AI techniques, reflecting the growing expectation that a prediction should come with an explanation of the features that drove it.

2.3 Explainable Machine Learning for Cardiovascular Risk

Napa et al. [5] presented a comparative analysis of explainable machine-learning models for cardiovascular risk stratification using clinical data and Shapley additive explanations (SHAP). Paul and Syed Masood [6] proposed an explainable machine-learning framework for cardiovascular risk prediction using structured health data. Both studies emphasise that, in a medical setting, a model's usefulness depends not only on its accuracy but also on the clarity with which its reasoning can be communicated. HealthScope follows this direction in a lightweight way: it uses inherently interpretable tree-based models and reports feature-importance scores for the selected Decision Tree models (Chapter 7).

2.4 Summary of Surveyed Studies

Table 2.1 – Surveyed studies and their relevance

Ref.	Authors (year)	Theme	Relevance to HealthScope
[1]	Maniruzzaman et al. (2020)	Classification and prediction of diabetes with a machine-learning paradigm	Comparing multiple classifiers for diabetes
[2]	Maniruzzaman et al. (2018)	Diabetes risk stratification; role of missing values and outliers	Motivates treating zeros as missing and imputing
[3]	Xie et al. (2019)	Risk prediction models for type 2 diabetes	Risk modelling from routine health variables
[4]	Thotad et al. (2023)	Diabetes detection on Indian demographic and health survey data	Survey-based diabetes classification
[5]	Napa et al. (2025)	Explainable ML for cardiovascular risk using SHAP	Explainability for cardiovascular models
[6]	Paul and Syed Masood (2026)	Explainable ML framework for cardiovascular risk on structured data	Structured-data cardiovascular modelling
[7]	Joshi and Dhakal (2021)	Type 2 diabetes prediction with logistic regression and ML	Logistic Regression as baseline
[8]	Lai et al. (2019)	Predictive models for diabetes mellitus	Predictive modelling for diabetes
[9]	Chang et al. (2023)	Pima Indians diabetes classification with ML algorithms	Same dataset as the diabetes module
[10]	Tasin et al. (2023)	Diabetes prediction with ML and explainable AI	Combining prediction with explanation

2.5 Research Gap and Contribution of HealthScope

Most of the surveyed studies concentrate on a single condition – predominantly diabetes – and present results as experiments rather than as an interactive tool. HealthScope contributes in three ways. First, it applies one consistent preprocessing, splitting and evaluation protocol to diabetes, heart disease and hypertension, so that the three conditions can be compared. Second, it connects the trained models to a Streamlit dashboard so that the analysis is usable beyond a notebook. Third, it is explicit about its limits: the models classify dataset-defined status, are evaluated only on held-out data from the same sources, and are presented as educational rather than diagnostic.

CHAPTER 3 – SYSTEM AND DATA REQUIREMENTS

3.1 Hardware Requirements

HealthScope is a software-based Python and Streamlit application. The datasets are small (the largest has 4,240 rows) and the models are classical machine-learning classifiers, so no GPU or specialised hardware is required. The following configuration is sufficient to run the project.

Table 3.1 – Hardware requirements

Resource	Minimum	Recommended
Processor	Dual-core 64-bit CPU	Quad-core 64-bit CPU
Memory (RAM)	4 GB	8 GB or more
Storage	1 GB free (libraries, datasets, model files)	2 GB free
Display	1366 × 768	1920 × 1080 for the dashboard
Network	Not required to run locally	Required only for installing packages and for deployment

3.2 Software Requirements

Table 3.2 – Software requirements

Software	Purpose
Python 3	Implementation language for data processing, model training and the dashboard
Streamlit	Web framework for the interactive dashboard (served locally at localhost:8501 during development)
Pandas, NumPy	Tabular data handling and numerical computation
scikit-learn	Imputation, scaling, classification algorithms and evaluation metrics
Joblib	Saving and loading trained models, preprocessing pipelines and evaluation results
Web browser	Viewing the Streamlit interface (any current desktop browser)
Operating system	Windows, macOS or Linux – all support the Python stack above

3.3 Functional Requirements

- The system shall let the user choose among Diabetes, Heart Disease and Hypertension assessments.
- The system shall collect the health measurements needed by the selected condition's model.
- The system shall return a risk label and an estimated risk value for the entered data.
- The system shall present visual comparisons of the entered values with reference ranges, together with key insights and suggested next steps.
- The system shall state clearly that the result is educational and is not a medical diagnosis.

3.4 Non-Functional Requirements

- **Usability:** a first-time user should be able to reach a result in a few steps without technical knowledge.
- **Reproducibility:** fixed random seeds (random_state = 42) and a stratified 80:20 split make the experiments repeatable.
- **Maintainability:** separate preprocessing, training, evaluation and interface stages (including an eda.py module) keep the code modular.
- **Responsiveness:** classical models predict near-instantly on a single record, which suits an interactive dashboard.

3.5 Dataset Inventory

HealthScope uses three disease-specific datasets. The exact survey or examination round is not specified in the available dataset information and has therefore not been assumed.

Table 3.3 – Dataset summary

Detail	Diabetes	Heart Disease	Hypertension
Dataset	Pima Indians Diabetes Database	UCI Heart Disease – Cleveland subset	Framingham Heart Study dataset
File	diabetes.csv	heart_disease.csv	hypertension.csv
Records	768	303	4,240
Total columns	9	14	16
Input features used	8	13	13
Target variable	Outcome	target	prevalentHyp
Task	Binary classification	Binary classification	Binary classification
Survey round	Not specified	Not applicable as a survey round; donated to UCI in 1988	Exact examination round not specified
Source	Kaggle (UCI ML)	UCI ML Repository	Kaggle

3.6 Target Class Distribution

Table 3.4 – Target class distribution

Dataset	Class 0 (negative)	Class 1 (positive)	Positive share
Diabetes (Outcome)	500 (65.1%)	268 (34.9%)	34.9%
Heart Disease (target)	138 (45.5%)	165 (54.5%)	54.5%
Hypertension (prevalentHyp)	2,923 (68.9%)	1,317 (31.1%)	31.1%

The diabetes and hypertension datasets are moderately imbalanced, with roughly one positive record for every two negative records. Accuracy alone can therefore be misleading, which is why precision, recall and F1-score are reported alongside it. The heart-disease dataset is close to balanced.

CHAPTER 4 – TECHNOLOGY STACK

HealthScope is built entirely on open-source Python technologies. The table below lists each layer of the stack and its role in the project.

Table 4.1 – Technology stack

Layer	Technology	Role in HealthScope
Programming language	Python	Data preparation, model training and dashboard logic
Data handling	Pandas, NumPy	CSV loading, tabular manipulation and numerical operations
Machine learning	scikit-learn	Preprocessing pipelines, classification algorithms and metrics
Preprocessing	SimpleImputer, StandardScaler	Median imputation of missing values and standardisation of features
Dashboard	Streamlit	Interactive web application and input widgets
Model persistence	Joblib	Saving and loading fitted models and evaluation artifacts
Visualisation	Dashboard charts	Risk indicator, BMI category and health-parameter comparisons

4.1 Python

Python is the primary language of data science and offers mature libraries for every stage of HealthScope. Its readability makes the project easy to maintain, and the same language is used for the analysis scripts and the web interface, avoiding a separate front-end technology.

4.2 Pandas and NumPy

Pandas provides the DataFrame structure used to load the three CSV files, inspect columns, count missing values and select feature and target columns. NumPy supplies the underlying n-dimensional arrays and numerical routines used by scikit-learn and by the dashboard when assembling a single-row input.

4.3 scikit-learn

scikit-learn [14] supplies the machine-learning components: SimpleImputer for missing-value handling, StandardScaler for feature standardisation, the LogisticRegression, DecisionTreeClassifier and RandomForestClassifier estimators, train-test splitting, and the metric functions (accuracy, precision, recall, F1-score and confusion matrix). Its consistent fit/predict interface makes it simple to train three algorithms with identical preprocessing.

4.4 Streamlit

Streamlit [15] turns Python scripts into interactive web applications. The HealthScope interface – landing page, disease selector, numeric input controls, result display and charts – is written in Python without separate HTML or JavaScript code, and runs locally on port 8501 during development.

4.5 Joblib

Joblib serialises Python objects efficiently. HealthScope uses it to store fitted models, preprocessing pipelines and evaluation results so that the dashboard can load a trained model instantly instead of retraining on every launch.

CHAPTER 5 – SYSTEM ARCHITECTURE

5.1 End-to-End Workflow

The high-level workflow is: healthcare CSV datasets → data inspection and preprocessing → feature preparation → classifier training and evaluation → model selection → saved model artifacts → Streamlit user interface → risk estimate and visual insights. An exploratory-analysis branch (class distribution, feature distributions and correlation analysis) runs alongside preprocessing and informs how the data are prepared.

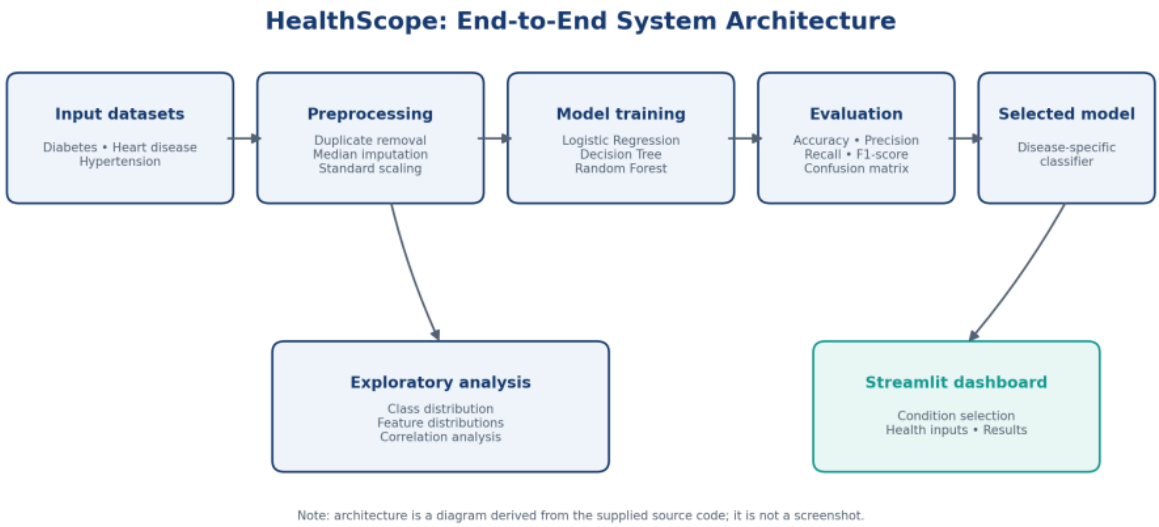


Figure 5.1 – HealthScope end-to-end system architecture

Input datasets are preprocessed, used to train and evaluate three classifiers, and the selected disease-specific model feeds the Streamlit dashboard. The diagram is derived from the source code; it is not a screenshot.

5.2 Component Responsibilities

Table 5.1 – Architecture components

Component	Responsibility	Input	Output
Dataset layer	Loads the diabetes, heart disease and hypertension CSV datasets	CSV records	Tabular datasets
Preprocessing layer	Handles missing values (median imputation) and standardises features	Feature columns	Transformed feature matrix
Exploratory analysis	Class distribution, feature distributions and correlation analysis (eda.py)	Cleaned datasets	Charts and statistical insight
Model layer	Trains Logistic Regression, Decision Tree and Random Forest	Transformed training features	Fitted classifiers
Evaluation layer	Compares models using accuracy, precision, recall, F1-score and confusion matrices	Test predictions and labels	Metric summaries; selected model
Presentation layer	Collects user inputs and displays the assessment and visual insights	User-entered health values	Educational risk result and suggestions

5.3 Design Principles

- **One pipeline per condition:** each disease has its own feature set, preprocessing and selected model, because the datasets differ in features, scales and target definitions.
- **Same protocol for every condition:** the same split, random seed, algorithms and metrics are used so that results are comparable.
- **Separation of training and serving:** models are trained and evaluated offline; the dashboard only loads artifacts and predicts.

- **Honest presentation:** the interface frames every output as educational and shows context (reference ranges, insights) rather than a bare number.

5.4 Data Flow During an Assessment

1. The user opens the dashboard and selects a condition on the Health Assessment page.
2. The page displays the input controls for that condition.
3. On pressing the assessment button, the entered values are assembled into a single record whose columns match the model's input features.
4. The record is passed through the saved preprocessing (imputation and scaling) and then to the selected classifier.
5. The predicted probability is converted into a risk label and displayed together with comparisons, key insights and suggested next steps.

CHAPTER 6 – DATA UNDERSTANDING AND PREPARATION

6.1 Diabetes Dataset – diabetes.csv

The Pima Indians Diabetes Database (obtained through Kaggle) contains **768 records** of women of Pima Indian heritage aged 21 years or older, originally collected by the National Institute of Diabetes and Digestive and Kidney Diseases [11], [12]. The file has nine columns: eight input features – Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction and Age – and the target **Outcome**, where 0 indicates no diabetes and 1 indicates diabetes. The target distribution is 500 negative and 268 positive records. The exact survey round is not identified in the dataset description.

6.2 Heart Disease Dataset – heart_disease.csv

The heart-disease data come from the **Cleveland subset** of the UCI Heart Disease dataset [13], [18], donated to the UCI repository in 1988. It contains **303 patients** with 13 input features – age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca and thal – and one target column. The original UCI target distinguishes disease severity from 0 to 4; the project uses a **binary** target in which 0 means absence and 1 means presence of heart disease. The binary distribution is 138 negative and 165 positive records.

6.3 Hypertension Dataset – hypertension.csv

The hypertension module uses the Framingham Heart Study dataset (obtained through Kaggle) [16], [17], which in its commonly distributed form has **4,240 records and 16 columns**. The target is **prevalentHyp**, where 0 indicates no prevalent hypertension and 1 indicates prevalent hypertension (2,923 negative and 1,317 positive records). Thirteen input features are used: male, age, currentSmoker, cigsPerDay, BPMeds, prevalentStroke, diabetes, totChol, sysBP, diaBP, BMI, heartRate and glucose. The original file also contains education and TenYearCHD, which are not among the 13 predictors. The exact examination round cannot be identified from the dataset description.

> **Interpretation note.** Because hypertension is itself defined by elevated blood pressure, using systolic and diastolic blood pressure (sysBP, diaBP) as inputs makes the task easier than predicting hypertension from indirect risk factors alone. The hypertension results in Chapter 10 should be read with this in mind (see Section 12.2).

6.4 Feature and Target Reference

Table 6.1 – Input features and target variables

Dataset	Input features	Target
Diabetes (8)	Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age	Outcome (0 = no diabetes, 1 = diabetes)
Heart Disease (13)	age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal	target (0 = absence, 1 = presence)
Hypertension (13)	male, age, currentSmoker, cigsPerDay, BPMeds, prevalentStroke, diabetes, totChol, sysBP, diaBP, BMI, heartRate, glucose	prevalentHyp (0 = no, 1 = prevalent hypertension)

Appendix A gives a short description of every feature. No separate feature-selection algorithm was applied: the configured input features are used directly for each disease-specific model.

6.5 Class Distribution

The figures below show the number of records in each class. They confirm the moderate imbalance of the diabetes and hypertension datasets and the near balance of the heart-disease dataset.

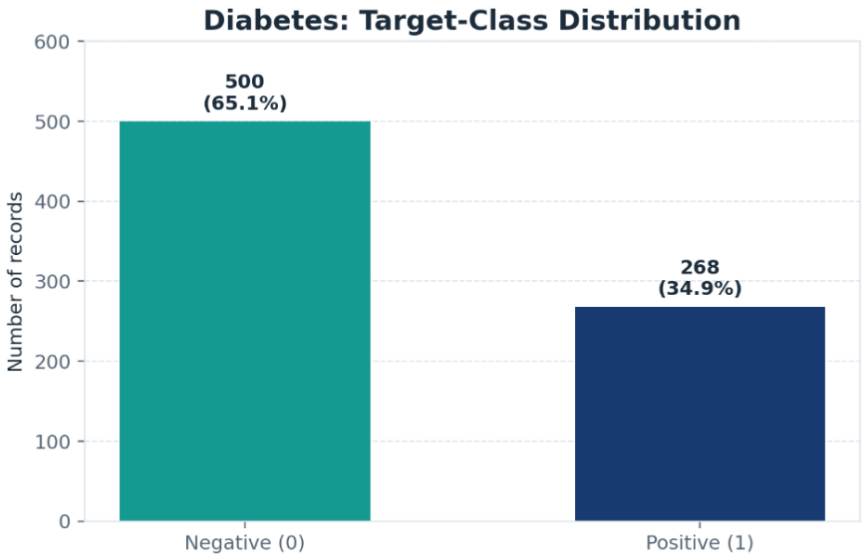


Figure 6.1 – Diabetes target-class distribution
500 negative (65.1%) and 268 positive (34.9%) records.

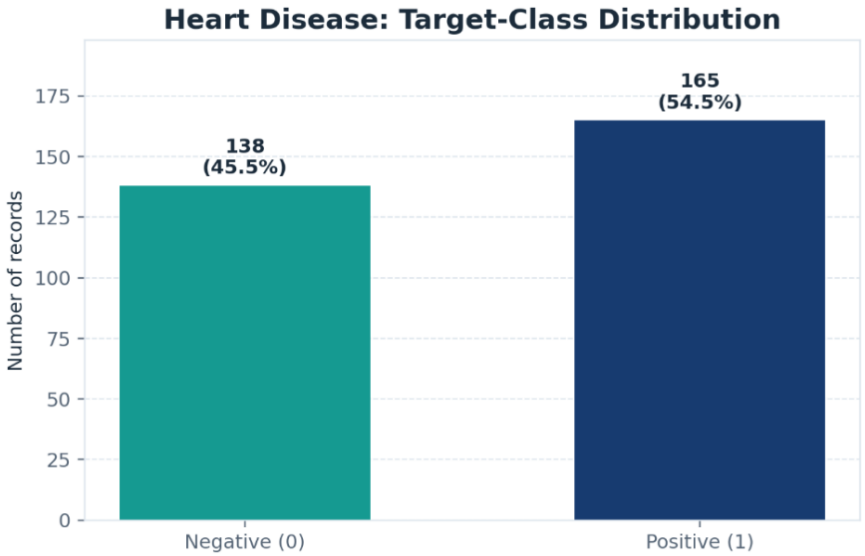


Figure 6.2 – Heart disease target-class distribution
138 negative (45.5%) and 165 positive (54.5%) records.

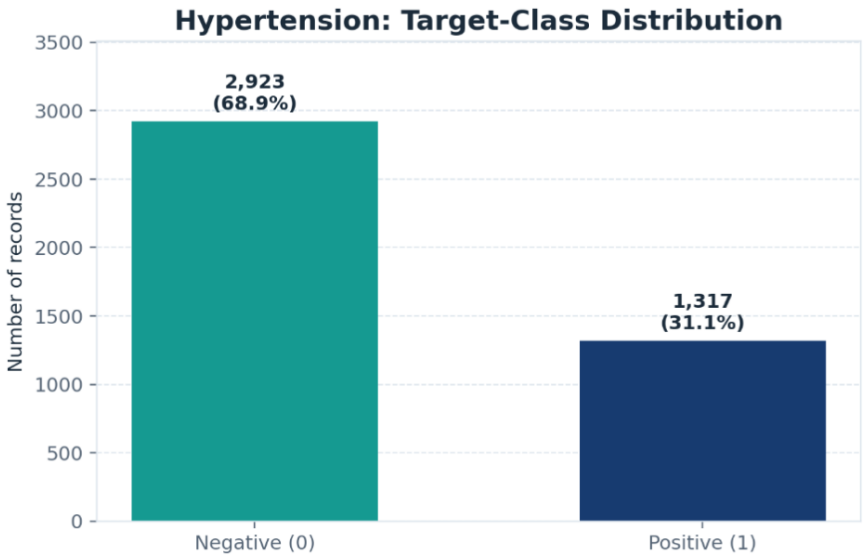


Figure 6.3 – Hypertension target-class distribution
2,923 negative (68.9%) and 1,317 positive (31.1%) records.

6.6 Data Preprocessing

6.6.1 Missing-value handling

SimpleImputer (median strategy) is used to fill missing values. The median is robust to the skewed distributions seen in several clinical variables, such as insulin and systolic blood pressure. The Framingham file is known to contain missing entries in some columns (for example glucose, cholesterol and BMI), so imputation matters most for the hypertension module.

6.6.2 Zero values in the diabetes dataset

In the Pima dataset, a value of zero for Glucose, BloodPressure, SkinThickness, Insulin or BMI is physiologically implausible and indicates a measurement that was not recorded. These zeros are therefore **treated as missing** and imputed rather than used as genuine measurements. The small spikes at zero visible in the Glucose and BMI histograms (Chapter 7) illustrate the issue.

6.6.3 Feature scaling

StandardScaler standardises every numerical feature to zero mean and unit variance. This is essential for Logistic Regression, whose optimisation is sensitive to feature scale, and harmless for the tree-based models. The imputer and scaler are combined in a scikit-learn pipeline so that exactly the same transformation is applied to training data, test data and a single record entered in the dashboard.

6.6.4 Train–test split

Each dataset is divided into **80% training and 20% testing data** using **stratified splitting** with **random_state = 42**. Stratification keeps the proportion of positive and negative records the same in both parts, which is important for the imbalanced datasets. The resulting test sets contain 154 records (diabetes), 61 records (heart disease) and 848 records (hypertension).

Table 6.2 – Train–test partition sizes

Dataset	Total records	Training (≈80%)	Test (≈20%)
Diabetes	768	614	154
Heart Disease	303	242	61
Hypertension	4,240	3,392	848

> **Methodological note.** The exact placement of the preprocessing steps relative to the train–test split, and whether any hyperparameter tuning was performed separately, should be confirmed from the source code before final submission.

> Fitting the imputer and scaler on the training portion only is the recommended practice because it prevents information from the test set leaking into training.

CHAPTER 7 – EXPLORATORY ANALYSIS AND FEATURE ENGINEERING

Exploratory data analysis (EDA) was carried out with the eda.py module to understand feature distributions, identify patterns and examine relationships between the health indicators and each target outcome. This chapter presents the target-class distributions (Chapter 6), selected feature distributions, Pearson correlation matrices and the feature-importance scores of the selected Decision Tree models.

7.1 Feature Distributions

7.1.1 Diabetes

The Glucose histogram is roughly bell-shaped with a centre near 100–120 mg/dL and a long right tail up to about 200. A small group of records sits at zero, which are the unrecorded measurements discussed in Section 6.6.2. The BMI histogram is centred near 30–35 with a similar small spike at zero. Age is strongly right-skewed: most participants are in their twenties and thirties, and the number of records falls steadily with age.

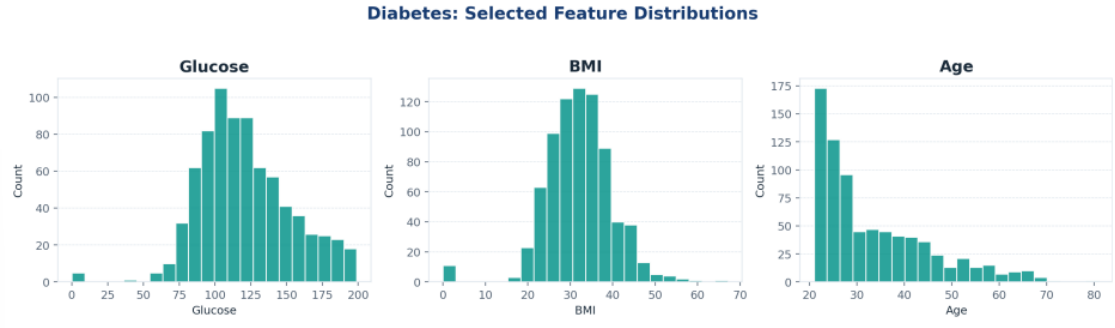


Figure 7.1 – Diabetes – selected feature distributions

Histograms of Glucose, BMI and Age. The bars at zero in Glucose and BMI correspond to missing measurements.

7.1.2 Heart disease

In the Cleveland data, age is spread between the thirties and the seventies with most patients between 40 and 65. Serum cholesterol (chol) is right-skewed, with most values between roughly 200 and 300 mg/dL and a few extreme values above 400. Maximum heart rate achieved (thalach) is left-skewed, concentrated between about 140 and 175 beats per minute, with fewer patients at lower rates.

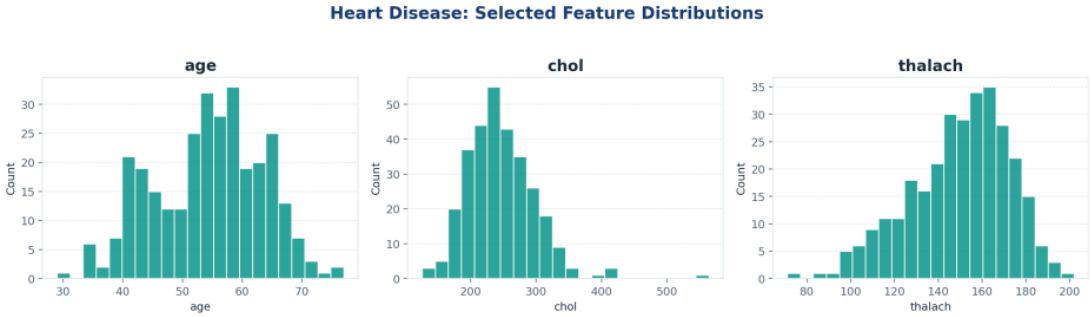


Figure 7.2 – Heart disease – selected feature distributions

Histograms of age, chol and thalach.

7.1.3 Hypertension

In the Framingham data, age ranges from the early thirties to about 70. Systolic blood pressure (sysBP) is right-skewed, with most values between roughly 100 and 150 mmHg and a small number of extreme readings reaching about 300. Diastolic blood pressure (diaBP) is closer to symmetric and centred near 80 mmHg. These extreme blood-pressure values are one reason scaling and a median-based imputer were preferred.

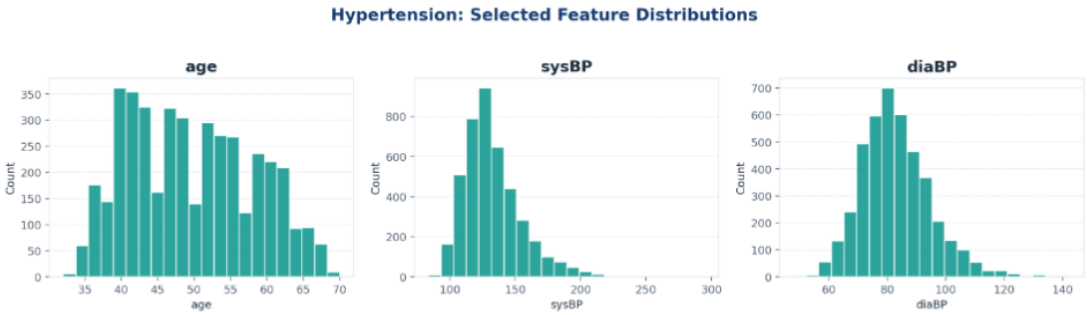


Figure 7.3 – Hypertension – selected feature distributions
Histograms of age, sysBP and diaBP.

7.2 Correlation Analysis

Pearson correlation matrices show the linear relationship between every pair of variables, including the target. Values close to +1 or −1 indicate strong positive or negative linear association; values near 0 indicate little linear relationship.

7.2.1 Diabetes

Glucose has the strongest correlation with the diabetes outcome (0.47), followed by BMI (0.29), Age (0.24) and Pregnancies (0.22). DiabetesPedigreeFunction (0.17) and Insulin (0.13) show weaker association, and BloodPressure (0.07) and SkinThickness (0.07) show almost none. Among the predictors, Age and Pregnancies are correlated (0.54), Insulin and SkinThickness are correlated (0.44), and BMI and SkinThickness are correlated (0.39), as expected from their physiological relationships.

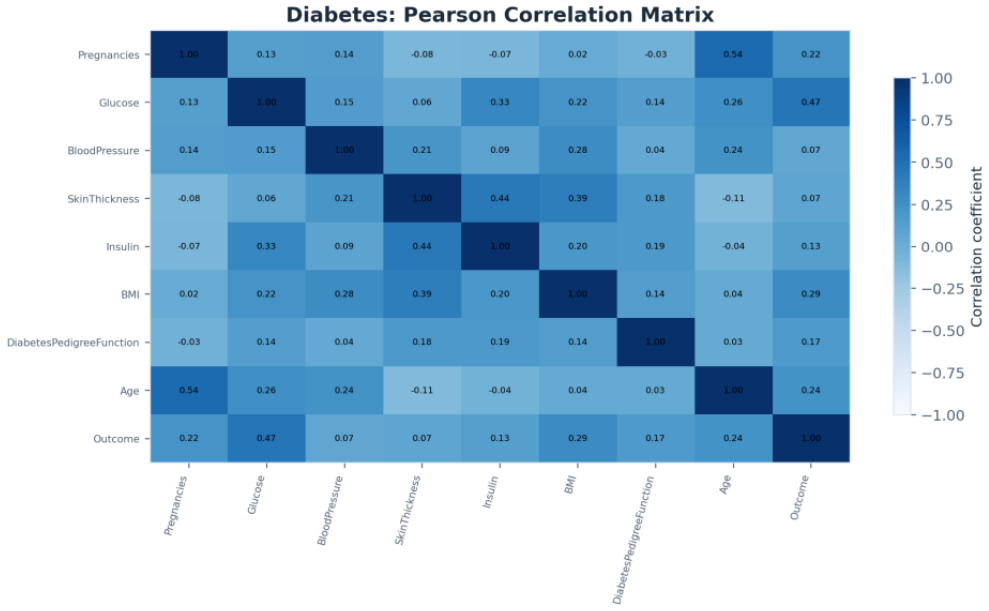


Figure 7.4 – Diabetes – Pearson correlation matrix
Glucose shows the strongest correlation with Outcome (0.47).

7.2.2 Heart disease

The heart-disease matrix shows moderate associations between the target and chest-pain type (cp), maximum heart rate (thalach), exercise-induced angina (exang), ST depression (oldpeak), the number of major vessels (ca) and thal. The relationship with thalach is negative: lower achievable heart rate accompanies disease presence, whereas the relationships with oldpeak, exang, ca and thal are positive. Cholesterol and fasting blood sugar show little direct linear association with the target in this dataset.

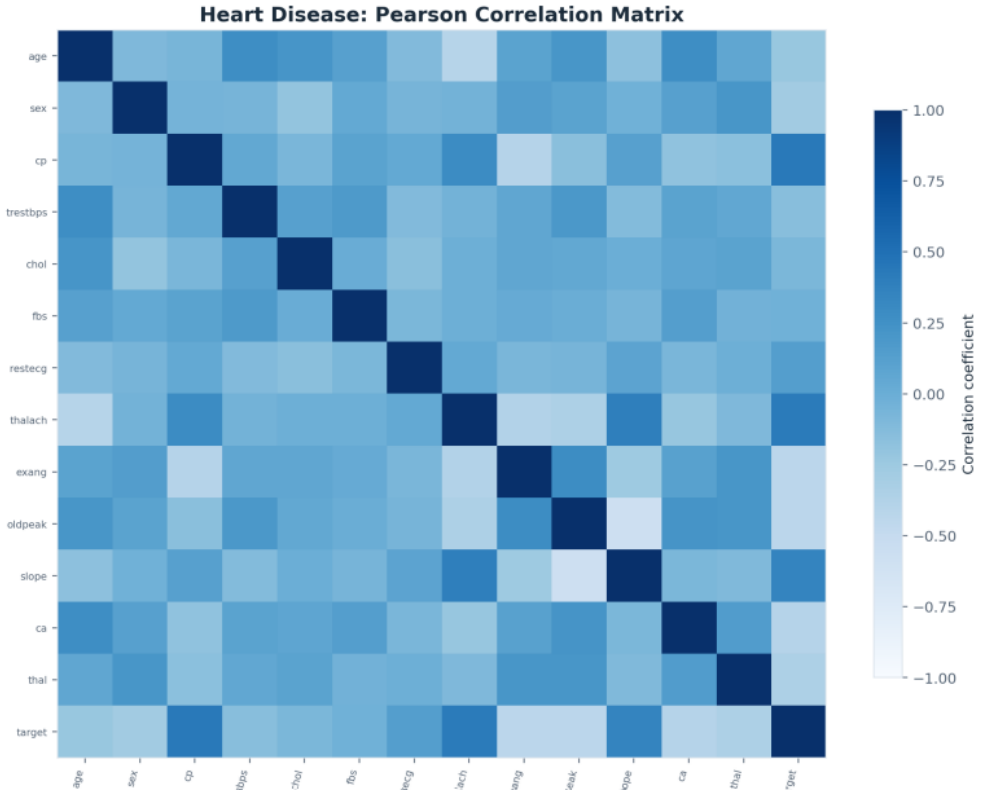


Figure 7.5 – Heart disease – Pearson correlation matrix

Correlations among the 13 input features and the target.

7.2.3 Hypertension

In the hypertension matrix, systolic and diastolic blood pressure show the strongest association with prevalentHyp, which is expected because hypertension is defined by blood-pressure level. Age and BPMeds (use of blood-pressure medication) show additional moderate association. Within the predictors, cigsPerDay is very strongly related to currentSmoker, sysBP is strongly related to diaBP, and the diabetes indicator is related to glucose.

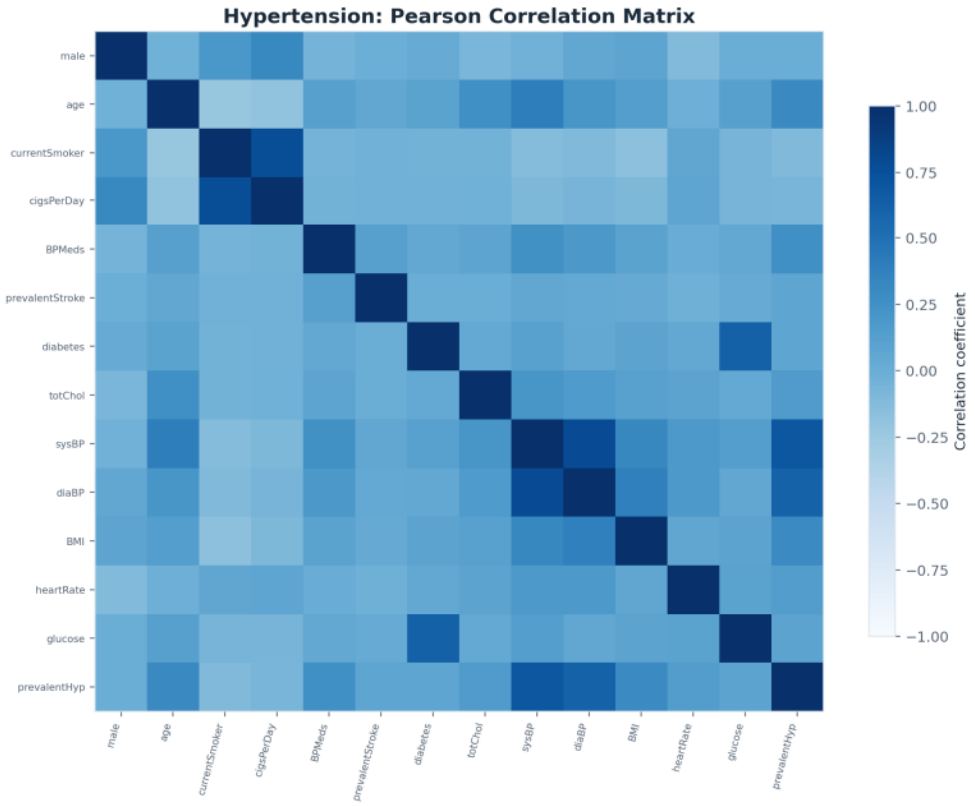


Figure 7.6 – Hypertension – Pearson correlation matrix
sysBP and diaBP show the strongest association with prevalentHyp.

7.3 Feature Engineering

No new derived features were created: each model uses the configured input columns of its dataset (8, 13 and 13 features respectively), and no separate feature-selection algorithm is applied. The feature engineering performed is therefore limited to transformations of existing columns – treating implausible zeros as missing, imputing, and standardising. This keeps the feature set aligned with the inputs the dashboard asks the user to provide.

7.4 Feature Importance of the Selected Models

The Decision Tree classifier assigns each feature an importance score based on how much it reduces impurity across the tree's splits. The scores are a useful, if approximate, explanation of what the selected model relies on.

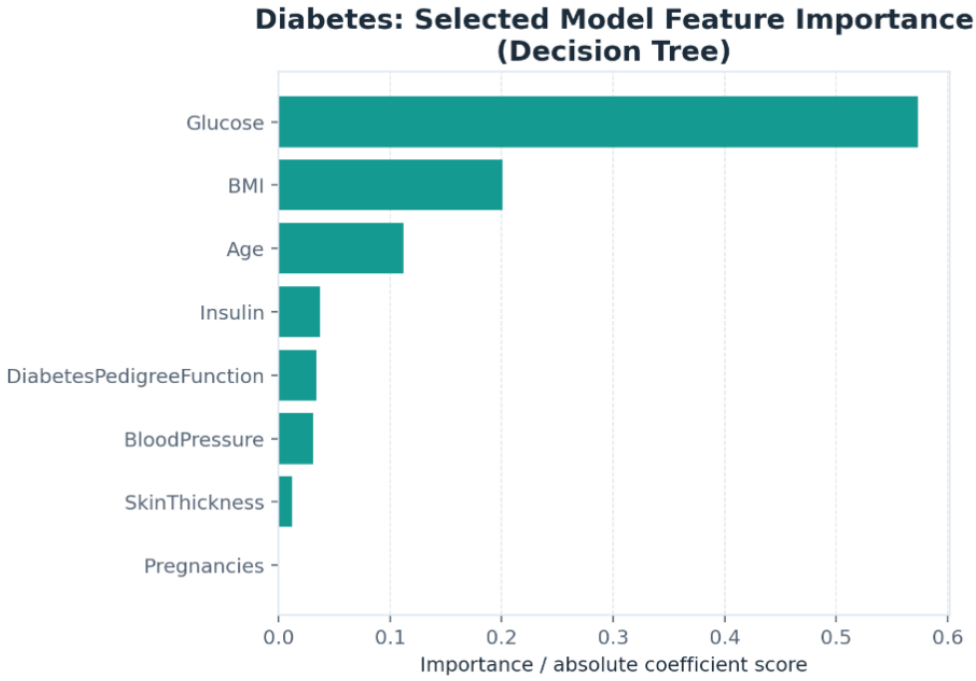


Figure 7.7 – Diabetes – feature importance (Decision Tree)
Glucose dominates, followed by BMI and Age.

For diabetes, **Glucose** carries by far the largest importance (about 0.57), followed by **BMI** (about 0.20) and **Age** (about 0.11). Insulin, DiabetesPedigreeFunction, BloodPressure and SkinThickness contribute little, and Pregnancies is effectively unused by the tree. This agrees with the correlation analysis, in which Glucose, BMI and Age were the three variables most strongly associated with the outcome.

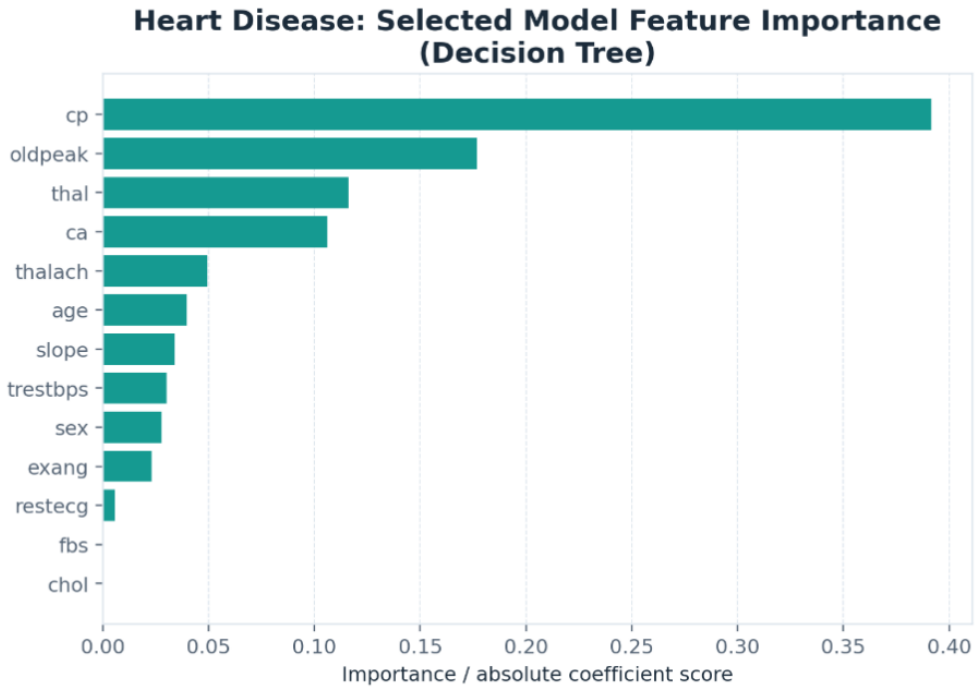


Figure 7.8 – Heart disease – feature importance (Decision Tree)
Chest-pain type (cp) is the most influential feature.

For heart disease, **cp** (chest-pain type) is the most important feature (about 0.39), followed by **oldpeak** (about 0.18), **thal** (about 0.12) and **ca** (about 0.11). thalach, age and slope contribute modestly, while chol and fbs contribute

almost nothing. A feature-importance chart for the hypertension model is not part of the available figures and is listed as future work.

> Decision-tree importance scores describe how the fitted model uses the features; they do not prove that a feature causes the condition. They can also be unstable when features are correlated or when the dataset is small, as with the 303-record heart-disease data.

CHAPTER 8 – MACHINE LEARNING METHODOLOGY

HealthScope follows a standard supervised-learning pipeline for each condition: preprocess the data, split it, train three classifiers, evaluate them on held-out data, and select the best one with a defined rule. This chapter describes the algorithms, their settings, the evaluation metrics and the selection rule.

8.1 Logistic Regression

Logistic Regression models the probability that a record belongs to the positive class as a logistic (sigmoid) function of a weighted sum of the input features. Training finds the weights that maximise the likelihood of the observed labels. Its main strengths are simplicity, speed and interpretable coefficients, and it outputs a probability that can be used directly as a risk estimate. Because it is linear, it cannot capture interactions or threshold effects unless these are engineered into the features. In HealthScope it serves as the **baseline** model, with max_iter = 1000 to guarantee convergence and random_state = 42.

8.2 Decision Tree

A Decision Tree learns a hierarchy of if–then rules. At each node the algorithm chooses the feature and threshold that best separate the classes, and it repeats this process for the resulting subsets until a stopping condition is reached. The model is easy to explain – a prediction corresponds to a path of readable conditions such as "glucose above a threshold" – and it naturally handles non-linear relationships and interactions. Its weakness is a tendency to overfit if allowed to grow without limit, so HealthScope restricts it with **max_depth = 5** and random_state = 42.

8.3 Random Forest

A Random Forest trains many decision trees on random bootstrap samples of the training data, each considering a random subset of features at every split, and combines their predictions by voting. Averaging many diverse trees reduces the variance of a single tree and usually improves stability. The trade-off is reduced interpretability compared with one tree. HealthScope uses **n_estimators = 100**, **max_depth = 6** and random_state = 42.

8.4 Hyperparameter Settings

Table 8.1 – Hyperparameter settings

Algorithm	Hyperparameter	Value
Logistic Regression	max_iter	1000
Logistic Regression	random_state	42
Decision Tree	max_depth	5
Decision Tree	random_state	42
Random Forest	n_estimators	100
Random Forest	max_depth	6
Random Forest	random_state	42

All other parameters keep the scikit-learn defaults. Fixed random seeds make every run reproducible. Whether hyperparameter tuning was carried out separately for each dataset should be confirmed from the source code.

8.5 Training Procedure

1. Load the disease-specific CSV file and separate the input features from the target column.
2. Treat implausible zeros in the diabetes features as missing values.
3. Split into 80% training and 20% test data with stratification and random_state = 42.
4. Apply median imputation and standard scaling through the preprocessing pipeline.
5. Train Logistic Regression, Decision Tree and Random Forest on the training portion.
6. Predict on the held-out test portion and compute accuracy, precision, recall, F1-score and the confusion matrix for each model.

7. Compute the weighted selection score, choose the best model and save the artifacts with Joblib for use by the dashboard.

8.6 Evaluation Metrics

All metrics are derived from the confusion matrix, written as [[TN, FP], [FN, TP]], where TN is true negatives, FP false positives, FN false negatives and TP true positives (the positive class is the presence of the condition).

Table 8.2 – Evaluation metrics

Metric	Formula	Interpretation in this project
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Share of all test records classified correctly; can be misleading when classes are imbalanced
Precision	TP / (TP + FP)	Of the records predicted to have the condition, the share that actually do
Recall (sensitivity)	TP / (TP + FN)	Of the records that truly have the condition, the share that are detected; the key metric for screening, because a miss is costly
F1-score	2 × Precision × Recall / (Precision + Recall)	Harmonic mean of precision and recall; balances false alarms and misses
Specificity	TN / (TN + FP)	Of the records without the condition, the share correctly cleared (reported as a derived measure)

ROC-AUC and calibration analysis were not part of the evaluation and are therefore not reported; they are listed as future work.

8.7 Model Selection Rule

The final model for each condition is chosen using a weighted score:

$$\textbf{Selection score} = 0.70 \times \textbf{F1-score} + 0.30 \times \textbf{Accuracy}$$

Weighting F1-score more heavily than accuracy favours models that find positive cases without producing too many false alarms, which is appropriate for the imbalanced diabetes and hypertension datasets. Based on the evaluation results, the Decision Tree was selected for diabetes and heart disease. Applying the same rule to the hypertension results also favours the Decision Tree (Chapter 10), and the final choice should be confirmed against the saved model-selection output of the application.

CHAPTER 9 – IMPLEMENTATION AND USER INTERFACE

9.1 Implementation Overview

HealthScope is implemented as a Python project with separate stages for data preparation, exploratory analysis (eda.py), model training and evaluation, and the Streamlit user interface. Trained models, preprocessing pipelines and evaluation results are saved with Joblib. The dashboard loads these artifacts, so no training happens while a user is interacting with the application. The complete code is available in the project's GitHub repository (Appendix D).

9.2 Application Pages

Table 9.1 – Dashboard pages and elements

Page / element	Purpose
Home	Landing page with the HealthScope branding, a Get Started action and three assessment cards (Diabetes, Heart Disease, Hypertension)
Health Assessment	Disease-selection dropdown and the numeric health-input form for the chosen condition, with an Assess My Risk button
About	Information about the project and its educational purpose
Result section	Risk label and estimated risk value, reference-range and BMI-category charts, health-parameter comparison, key insights and suggested next steps

9.3 Landing Page

The landing page is branded HealthScope and includes Home, Health Assessment and About navigation, a Get Started action and three assessment cards for diabetes, heart disease and hypertension. The page runs locally at localhost:8501 during development.

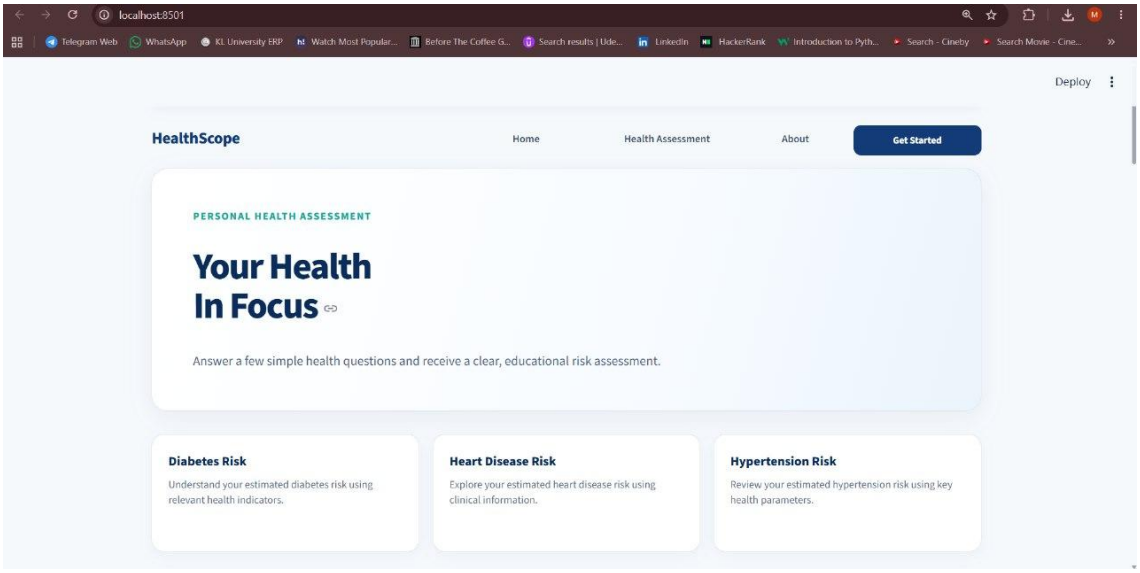


Figure 9.1 – HealthScope landing page

Landing page with navigation and entry points for the three health assessment categories.

9.4 Health Assessment Form

The assessment page includes a disease-selection dropdown that exposes the Diabetes, Heart Disease and Hypertension choices, followed by numeric health-input controls. The diabetes form collects age, blood glucose, blood pressure, BMI, number of pregnancies, a recent diabetes-test checkbox, skin thickness, insulin and a diabetes family-history score. The user submits the form with the **Assess My Risk** button.



Figure 9.2 – Disease assessment selector

The selector exposes Diabetes, Heart Disease and Hypertension assessment choices.

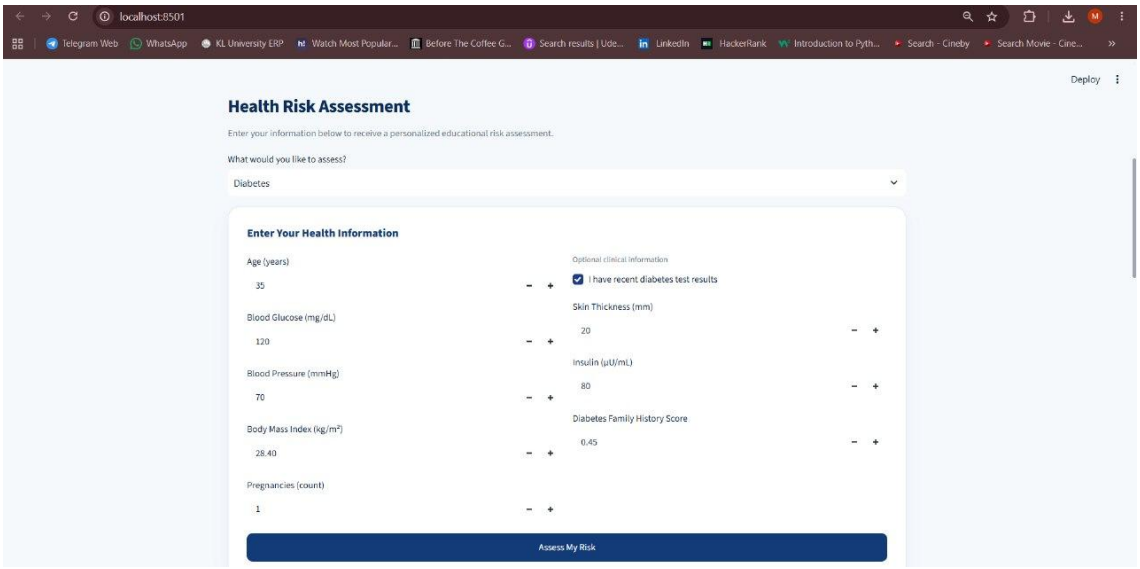


Figure 9.3 – Diabetes assessment input form

The form collects demographic and clinical values for an individual diabetes risk assessment.

The heart-disease and hypertension modules use the same flow, with inputs corresponding to the 13 features listed in Table 6.1 for each model. Screenshots of those two forms were not available for this report.

9.5 Assessment Result

After submission, the page displays a risk label and an estimated risk probability. In the supplied example, the result is labelled **Moderate Risk** with an estimated risk probability of **51.7%**. This value is the output of the model for one individual's example inputs; it is **not** a measure of model accuracy, which is reported separately in Chapter 10.

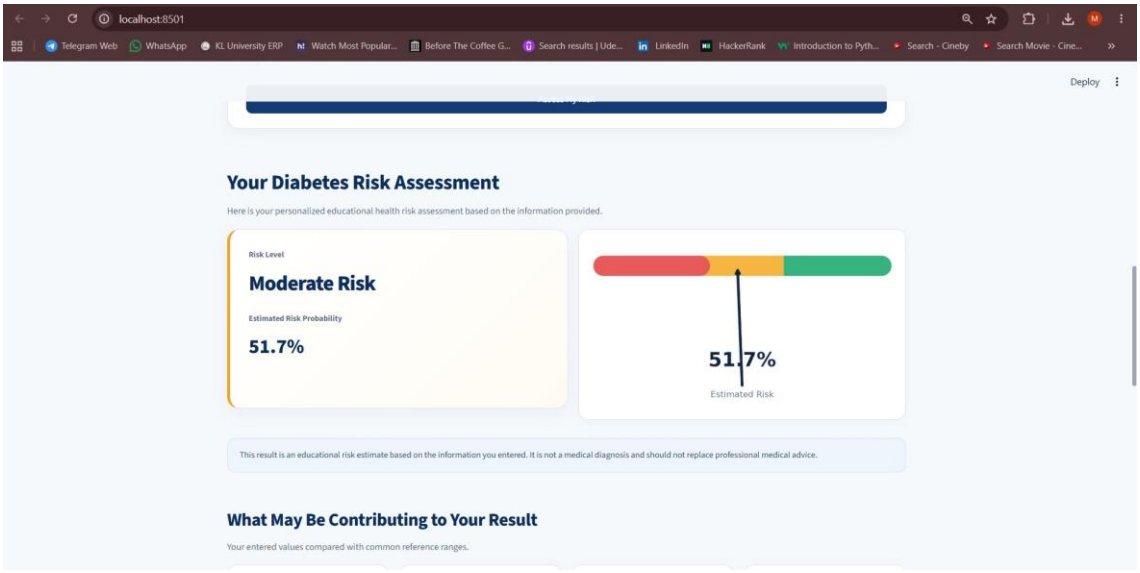


Figure 9.4 – Example diabetes risk result

The displayed example is a 51.7% estimated risk, labelled Moderate Risk.

9.6 Visual Insights and Suggested Next Steps

Below the result, the page compares the user's inputs with reference ranges: a glucose reference-range graphic, a BMI-category ring chart and a relative health-parameter comparison chart. A key-insights section summarises the factors that contribute to the result, and a suggested-next-steps section gives general educational advice. The interface states explicitly that the assessment is educational and is not a medical diagnosis.

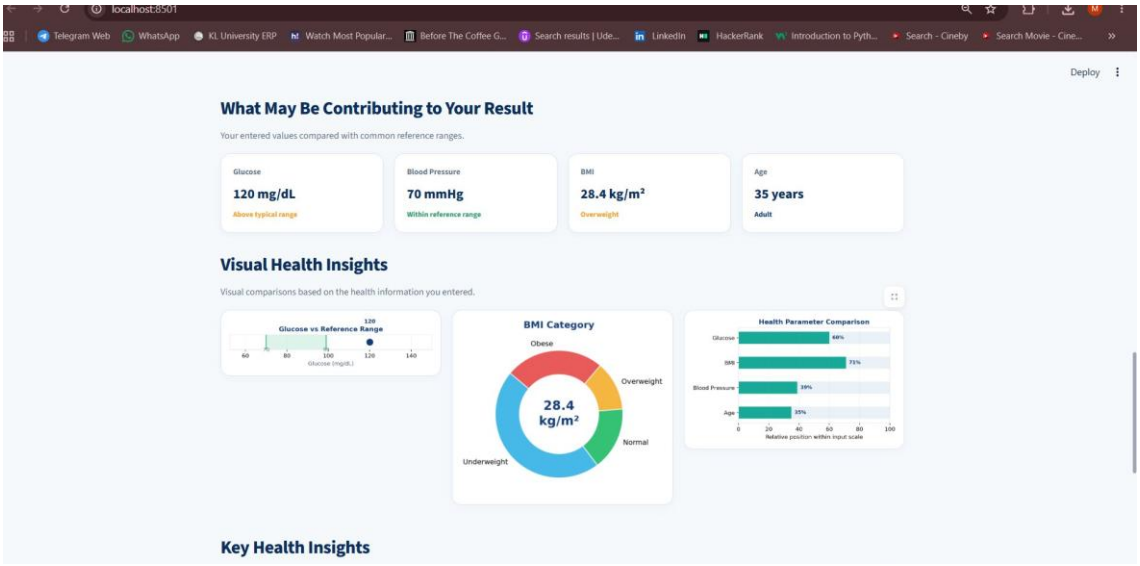


Figure 9.5 – Visual health insights

Individual input values are compared with reference ranges and shown through charts.

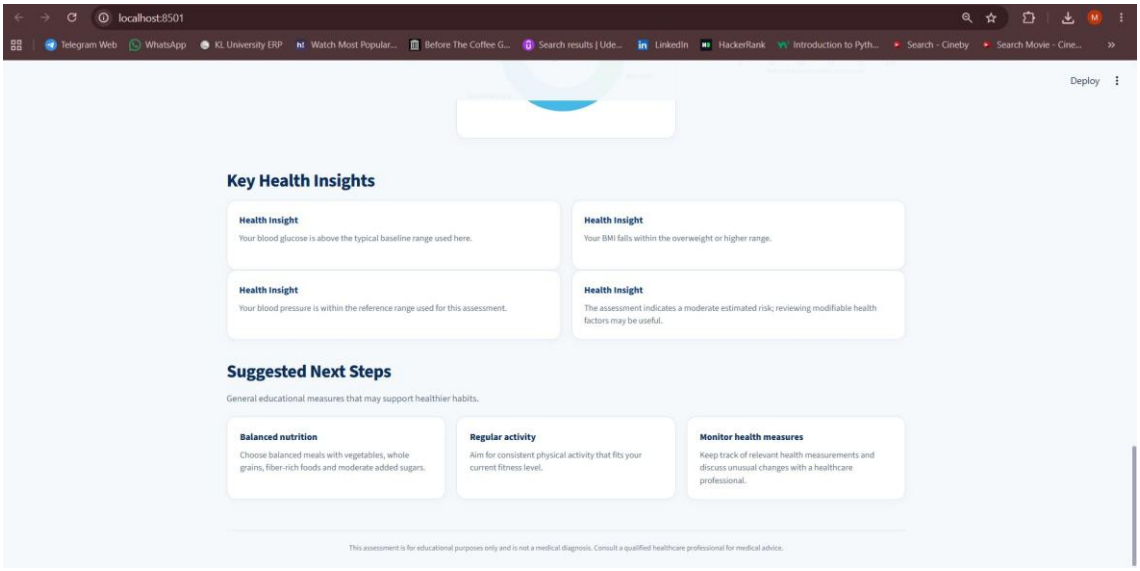


Figure 9.6 – Key insights and suggested next steps

The page summarises contributing factors and presents general educational suggestions.

> These visuals are application-level explanations for one individual's entered values. They should not be confused with the dataset-wide EDA charts of Chapter 7.

9.7 User Journey

1. Open HealthScope and read the landing page.
2. Choose Get Started or open Health Assessment from the navigation.
3. Select the condition to assess.
4. Enter the requested health measurements.
5. Press Assess My Risk.
6. Review the risk label, the comparison charts, the key insights and the suggested next steps, and consult a healthcare professional for any concern.

CHAPTER 10 – RESULTS AND PERFORMANCE ANALYSIS

The models were evaluated on held-out test sets created with an 80:20 stratified split and random_state = 42. Accuracy, precision, recall, F1-score and confusion matrices were used to compare Logistic Regression, Decision Tree and Random Forest. Metric values are those obtained in the project's evaluation; the derived measures in Section 10.4 were calculated from the reported confusion matrices.

10.1 Diabetes Prediction

The test set contains 154 records (100 negative and 54 positive).

Table 10.1 – Diabetes – test-set performance (154 records)

Algorithm	Accuracy	Precision	Recall	F1-score
Logistic Regression	70.78%	60.00%	50.00%	54.55%
Decision Tree	75.97%	63.93%	72.22%	67.83%
Random Forest	74.03%	66.67%	51.85%	58.33%

Confusion matrices in the form [[TN, FP], [FN, TP]]: Logistic Regression [[82, 18], [27, 27]]; Decision Tree [[78, 22], [15, 39]]; Random Forest [[86, 14], [26, 28]]. The Decision Tree achieved the highest accuracy (75.97%) and F1-score (67.83%) and was selected as the diabetes classifier. Its recall of 72.22% is much higher than that of Logistic Regression (50.00%) and Random Forest (51.85%): it misses 15 diabetic cases, compared with 27 and 26, at the cost of more false alarms (22, compared with 18 and 14).

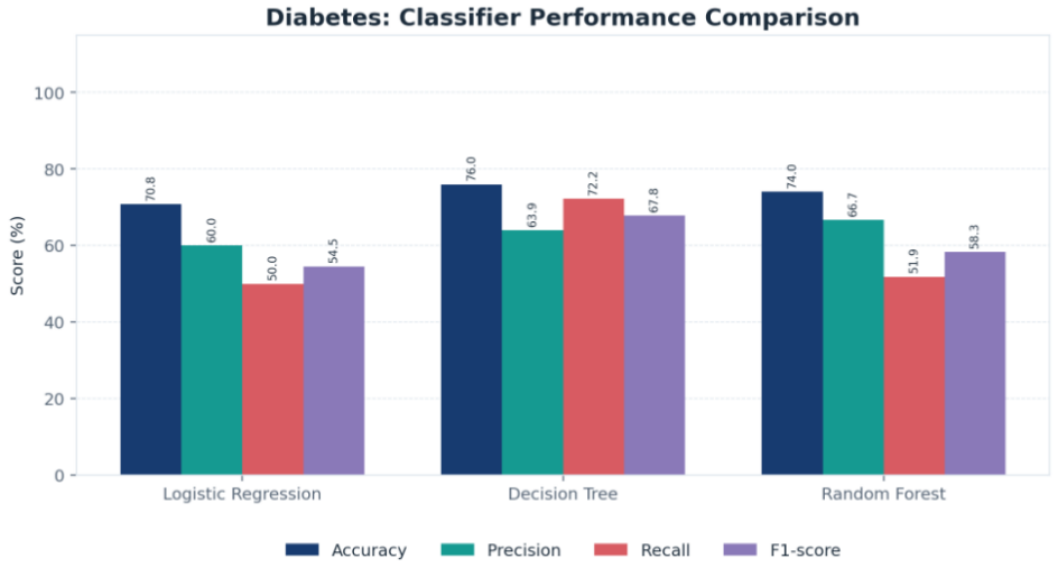


Figure 10.1 – Diabetes – classifier performance comparison
Accuracy, precision, recall and F1-score of the three classifiers.

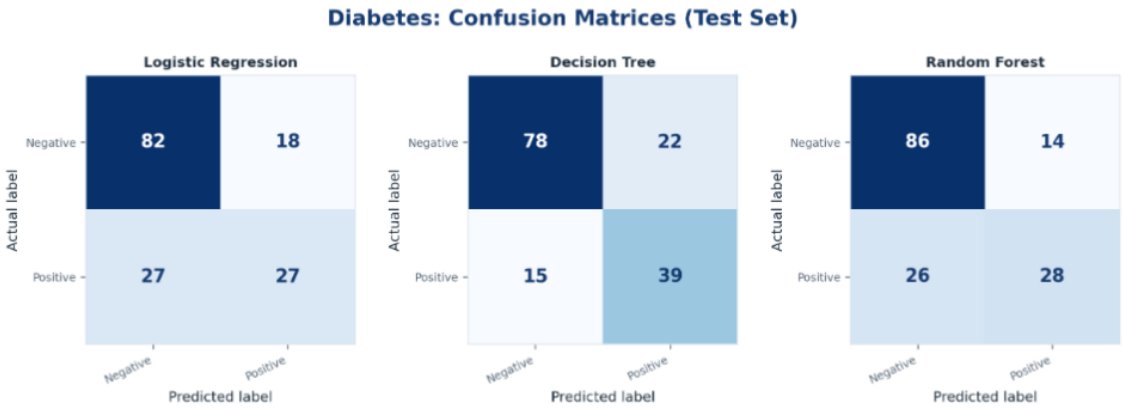


Figure 10.2 – Diabetes – confusion matrices (test set)

Confusion matrices of Logistic Regression, Decision Tree and Random Forest on 154 test records.

10.2 Heart Disease Prediction

The test set contains 61 records (28 negative and 33 positive).

Table 10.2 – Heart disease – test-set performance (61 records)

Algorithm	Accuracy	Precision	Recall	F1-score
Logistic Regression	78.69%	76.32%	87.88%	81.69%
Decision Tree	81.97%	75.00%	100.00%	85.71%
Random Forest	81.97%	77.50%	93.94%	84.93%

Confusion matrices: Logistic Regression [[19, 9], [4, 29]]; Decision Tree [[17, 11], [0, 33]]; Random Forest [[19, 9], [2, 31]]. The Decision Tree achieved the highest F1-score (85.71%) and tied with Random Forest for the highest accuracy (81.97%), and was selected as the heart-disease classifier. It detected all 33 positive cases (100% recall) but produced 11 false positives, the most of the three models, so its precision (75.00%) is slightly lower than that of Random Forest (77.50%).

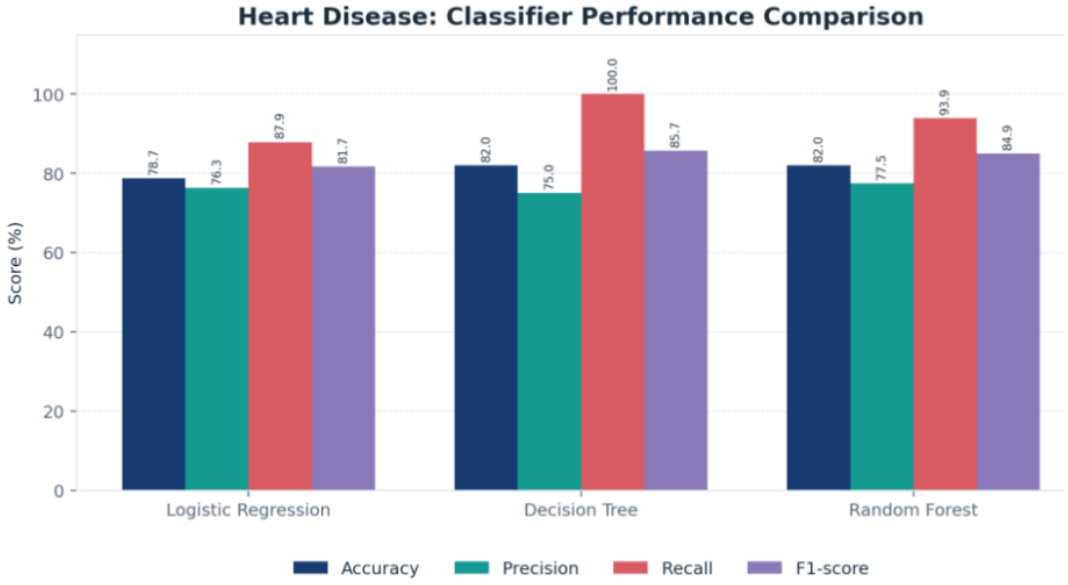


Figure 10.3 – Heart disease – classifier performance comparison

Accuracy, precision, recall and F1-score of the three classifiers.

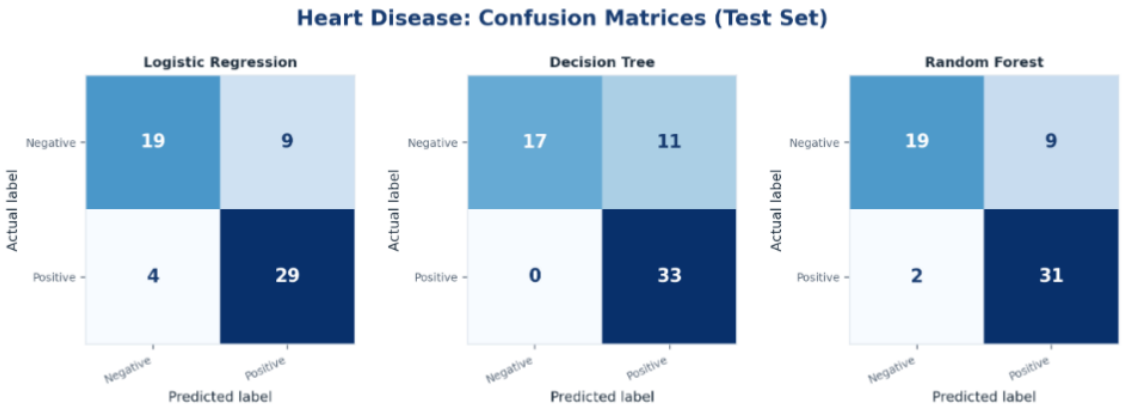


Figure 10.4 – Heart disease – confusion matrices (test set)
Confusion matrices on 61 test records.

10.3 Hypertension Prediction

The test set contains 848 records (585 negative and 263 positive). The metrics below were read from the hypertension evaluation figures and are consistent with the confusion matrices shown beside them.

Table 10.3 – Hypertension – test-set performance (848 records)

Algorithm	Accuracy	Precision	Recall	F1-score
Logistic Regression	87.85%	82.79%	76.81%	79.68%
Decision Tree	89.39%	80.14%	87.45%	83.64%
Random Forest	89.39%	80.78%	86.31%	83.46%

Confusion matrices: Logistic Regression [[543, 42], [61, 202]]; Decision Tree [[528, 57], [33, 230]]; Random Forest [[531, 54], [36, 227]]. The Decision Tree and Random Forest tie for the highest accuracy (89.39%); the Decision Tree has the higher F1-score (83.64% versus 83.46%) and the higher recall (87.45% versus 86.31%), while Random Forest has slightly higher precision (80.78% versus 80.14%). Logistic Regression has the highest precision of all (82.79%) but misses the most positive cases (61).

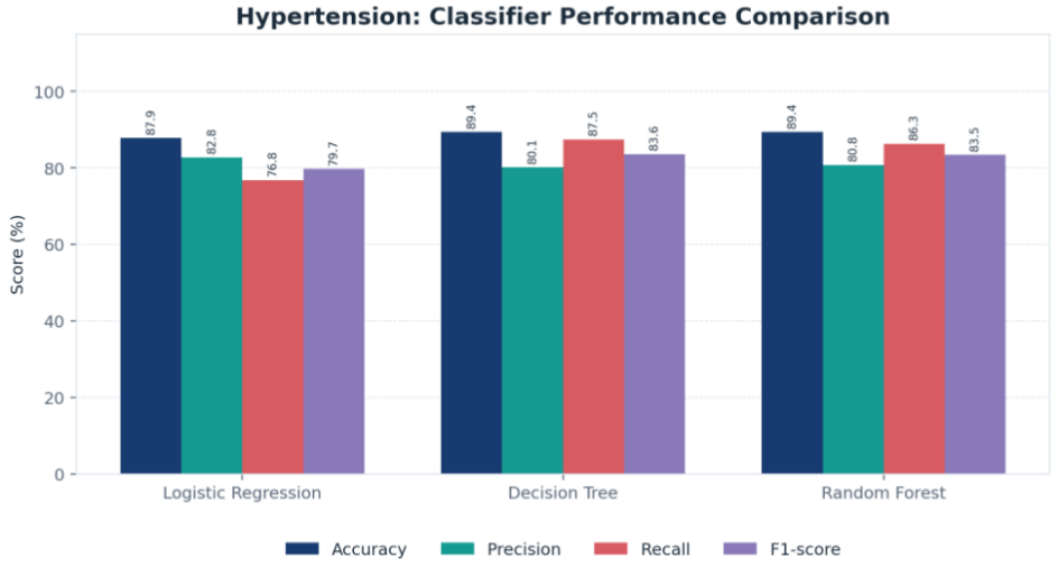


Figure 10.5 – Hypertension – classifier performance comparison
Accuracy, precision, recall and F1-score of the three classifiers.

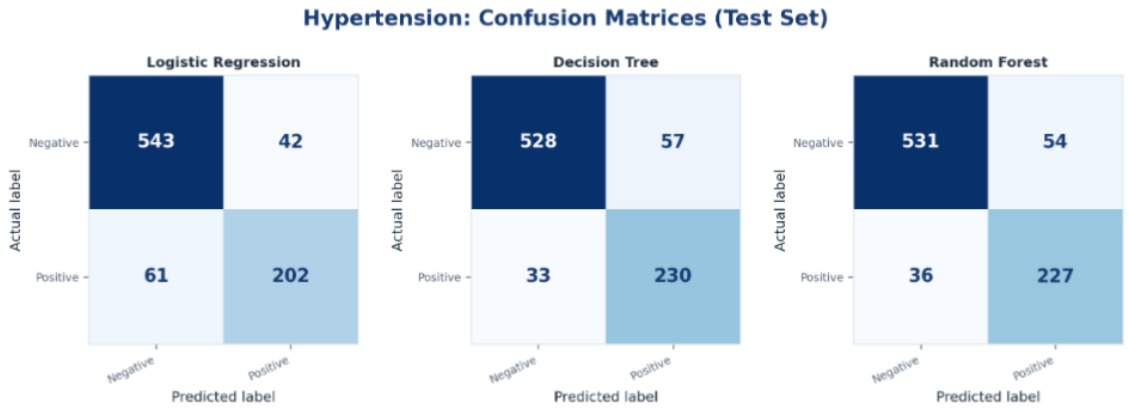


Figure 10.6 – Hypertension – confusion matrices (test set)
Confusion matrices on 848 test records.

> **Verification status.** The hypertension evaluation was described as requiring verification in the project's implementation summary. The numbers above agree with the evaluation figures and with the arithmetic of the confusion matrices, but the team should re-run the hypertension pipeline from the source code and confirm the final selected model before submission.

10.4 Derived Measures from the Confusion Matrices

Table 10.4 – Confusion-matrix counts and derived measures

Condition	Algorithm	TN	FP	FN	TP	Specificity	Miss rate (FN rate)
Diabetes	Logistic Regression	82	18	27	27	82.00%	50.00%
	Decision Tree	78	22	15	39	78.00%	27.78%
	Random Forest	86	14	26	28	86.00%	48.15%
Heart Disease	Logistic Regression	19	9	4	29	67.86%	12.12%
	Decision Tree	17	11	0	33	60.71%	0.00%
	Random Forest	19	9	2	31	67.86%	6.06%
Hypertension	Logistic Regression	543	42	61	202	92.82%	23.19%
	Decision Tree	528	57	33	230	90.26%	12.55%
	Random Forest	531	54	36	227	90.77%	13.69%

The miss rate is the share of truly positive records that the model failed to detect (100% − recall). For a screening-style tool a low miss rate is desirable, but it normally comes with a lower specificity, meaning more people are flagged who do not have the condition. The Decision Tree models trade some specificity for higher recall in all three conditions: for diabetes (specificity 78.00%), heart disease (60.71%) and hypertension (90.26%).

10.5 Model Selection Scores

Table 10.5 – Weighted model-selection scores

Condition	Algorithm	F1-score	Accuracy	Score (0.7·F1 + 0.3·Acc)	Outcome
Diabetes	Logistic Regression	54.55%	70.78%	59.42	
	Decision Tree	67.83%	75.97%	70.27	Selected
	Random Forest	58.33%	74.03%	63.04	
Heart Disease	Logistic Regression	81.69%	78.69%	80.79	
	Decision Tree	85.71%	81.97%	84.59	Selected
	Random Forest	84.93%	81.97%	84.04	
Hypertension	Logistic Regression	79.68%	87.85%	82.14	
	Decision Tree	83.64%	89.39%	85.36	Selected
	Random Forest	83.46%	89.39%	85.24	

The rule selects the Decision Tree for every condition. The margin over Random Forest is substantial for diabetes (70.27 versus 63.04) and heart disease (84.59 versus 84.04) and very small for hypertension (85.37 versus 85.24).

10.6 Cross-Condition Comparison

Table 10.6 – Selected model for each condition

Condition	Test records	Selected model	Accuracy	Precision	Recall	F1-score
Diabetes	154	Decision Tree	75.97%	63.93%	72.22%	67.83%
Heart Disease	61	Decision Tree	81.97%	75.00%	100.00%	85.71%
Hypertension	848	Decision Tree	89.39%	80.14%	87.45%	83.64%

Diabetes is the hardest of the three tasks (lowest accuracy and F1-score), which is consistent with the modest correlations between its features and the outcome (Chapter 7) and with the well-known difficulty of the Pima dataset. Hypertension reaches the highest accuracy, but largely because blood-pressure readings are among its inputs. Heart disease reaches the highest F1-score, but on the smallest test set.

10.7 Discussion

The Decision Tree classifier performed best according to the project's model-selection rule for all three conditions. It achieved 75.97% accuracy for diabetes, 81.97% for heart disease and 89.39% for hypertension, with F1-scores of 67.83%, 85.71% and 83.64% respectively. A likely reason for its advantage is that the depth limit of 5 allows it to capture threshold effects in clinical variables (for example, glucose above a level) without overfitting small datasets, while it remains easy to explain.

These results apply only to the available held-out test sets. In particular, the 100% recall obtained for heart disease comes from 33 positive test cases and does not establish perfect performance on unseen populations. With only 61 test records, a difference of one or two predictions changes the percentages noticeably. Independent validation is needed before any clinical use.

Logistic Regression, the linear baseline, had the highest precision for hypertension but the lowest recall in diabetes and hypertension, and lower recall than both tree-based models for heart disease. Random Forest, an ensemble, gave results close to the Decision Tree for heart disease and hypertension but did not improve on it, suggesting that with these small tabular datasets the extra model complexity brings no clear gain.

CHAPTER 11 – TESTING AND VALIDATION

11.1 Testing Approach

Two kinds of evidence are available. First, **model-level hold-out evaluation** with a stratified 20% test set for each condition, summarised by classification metrics and confusion matrices (Chapter 10). Second, **functional walkthrough testing** of the Streamlit interface, demonstrated by screenshots of a complete user journey from the landing page to the input form, result, charts and suggested next steps (Chapter 9).

11.2 Test Cases with Available Evidence

Table 11.1 – Test cases with available evidence

Test ID	Test description	Expected result	Evidence
TC-01	Open the application home page	HealthScope landing page is displayed	Figure 9.1
TC-02	Open the health assessment form	Disease selector and input fields are displayed	Figures 9.2 and 9.3
TC-03	Submit example diabetes inputs	Risk estimate and risk category are displayed	Figure 9.4
TC-04	Review health insights	Reference comparisons, charts and key insights are displayed	Figures 9.5 and 9.6
TC-05	Check the educational disclaimer	Result states that it is educational and not a diagnosis	Figures 9.4 to 9.6
TC-06	Train and compare classifiers – diabetes	Metrics and confusion matrices produced for three models	Tables 10.1, 10.4; Figures 10.1, 10.2
TC-07	Train and compare classifiers – heart disease	Metrics and confusion matrices produced for three models	Tables 10.2, 10.4; Figures 10.3, 10.4
TC-08	Train and compare classifiers – hypertension	Metrics and confusion matrices produced for three models	Tables 10.3, 10.4; Figures 10.5, 10.6
TC-09	Metric consistency check	Reported metrics agree with confusion-matrix arithmetic	Verified for all nine model results (Section 10.4)

11.3 Planned Tests

The following tests have not yet been carried out and are planned as part of future work.

Table 11.2 – Planned tests

Test area	Description	Status
Unit tests	Preprocessing functions (zero-as-missing, imputation, scaling) and input assembly	Planned
Input validation	Out-of-range, negative and missing values entered in the dashboard	Planned
Heart disease and hypertension flows	End-to-end assessment screenshots and results for both conditions	Planned
Cross-browser and device checks	Rendering on several browsers and screen sizes	Planned
Performance and load	Response time with several simultaneous users	Planned
External validation	Evaluation on independent datasets and subgroups	Planned
Security and privacy	Handling of user-entered health data in a deployed setting	Planned

11.4 Validation Status

The system has been validated only on held-out portions of the same three datasets used for training. Independent validation with external data, clinical validation and production-readiness checks have not been established and should not be assumed.

CHAPTER 12 – CHALLENGES AND LIMITATIONS

12.1 Challenges Encountered

- **Heterogeneous datasets:** the three datasets have different feature names, scales and target definitions, so each condition needs its own feature mapping and preprocessing pipeline.
- **Missing and implausible values:** the diabetes data contain zeros that represent unrecorded measurements, and the hypertension data contain missing entries in several columns. These needed explicit treatment through zero-as-missing handling and median imputation.
- **Class imbalance:** the diabetes and hypertension datasets have roughly one positive record per two negatives, so accuracy had to be interpreted together with precision, recall and F1-score, and stratified splitting was used.
- **Very small test set for heart disease:** the 61-record test set makes its metrics sensitive to a handful of predictions.
- **Balancing interpretability and performance:** shallow trees and a linear baseline were preferred because they can be explained, even though more complex models might fit slightly better.
- **Connecting analysis to an interface:** persisting preprocessing and models so that the dashboard applies exactly the same transformations as training required careful pipeline design.

12.2 Limitations

- **Dataset-defined status, not future risk:** the models classify disease status as labelled in each dataset. They do not predict future onset and do not provide clinically validated individual risk probabilities.
- **Blood-pressure inputs for hypertension:** the hypertension model uses sysBP, diaBP and BPMeds as inputs, and hypertension is defined by blood pressure. Its high accuracy therefore partly reflects the closeness between inputs and label and should not be read as the ability to detect hypertension from independent risk factors.
- **Limited external validity:** the evaluation uses held-out data from the same sources. The Pima data describe one population (women of Pima Indian heritage aged 21 or older), the Cleveland data describe one clinic's patients, and the Framingham data describe one long-running cohort. Performance on other populations is unknown.
- **Small sample for heart disease:** 303 records in total and 61 test records limit the strength of conclusions; the 100% recall must be interpreted cautiously.
- **No probability calibration or ROC-AUC:** the displayed risk value is an educational estimate and may not be a calibrated clinical probability.
- **Hypertension verification:** the hypertension evaluation and final model selection should be re-confirmed from the source code, and no feature-importance chart is available for that module.
- **Dashboard evidence:** screenshots demonstrate one diabetes assessment; heart-disease and hypertension assessment outputs have not been documented.
- **Deployment:** public deployment, authentication, privacy controls and production hosting have not been confirmed.
- **Fairness:** subgroup performance by sex, age or other characteristics has not been assessed.

12.3 Ethical and Safety Considerations

Health-related tools carry a responsibility to avoid harm. HealthScope is presented as an educational prototype: it labels every result as educational, encourages consultation with a healthcare professional, and avoids making diagnostic claims. Any future deployment that handles real user data would need informed consent, data-protection measures, clear communication of uncertainty, and review by clinical experts.

CHAPTER 13 – CONCLUSION AND FUTURE SCOPE

13.1 Conclusion

HealthScope demonstrates a complete data-science workflow that connects public healthcare datasets, preprocessing, exploratory analysis, classification models, model evaluation and an interactive Streamlit interface. It addresses three conditions – diabetes, heart disease and hypertension – with a consistent protocol: median imputation, standard scaling, a stratified 80:20 split with random_state = 42, three classifiers with fixed hyperparameters, and a selection rule weighting F1-score at 70% and accuracy at 30%.

On the held-out test sets, the Decision Tree was the best-performing model for every condition, with accuracies of 75.97% (diabetes), 81.97% (heart disease) and 89.39% (hypertension) and F1-scores of 67.83%, 85.71% and 83.64%. Exploratory analysis and feature-importance scores showed that the models rely on clinically sensible variables – glucose, BMI and age for diabetes; chest-pain type, ST depression, thal and ca for heart disease; and blood pressure for hypertension. The dashboard presents an accessible way to enter health values and review an educational estimate with visual context and suggested next steps.

The project also documents its boundaries honestly: it classifies dataset-defined status, has not been externally validated, and is not a diagnostic device. These boundaries define the most valuable next steps.

13.2 Status Summary

Table 13.1 – Implementation status

Component	Status	Description
Dataset integration	Completed	Diabetes, heart disease and hypertension datasets are incorporated
Data preprocessing	Completed	Missing-value handling and feature scaling; zeros treated as missing for diabetes
Exploratory data analysis	Completed	eda.py module; distributions, correlations and class balance
Machine-learning models	Completed	Logistic Regression, Decision Tree and Random Forest trained and compared
Model evaluation	Completed for all three conditions; hypertension to be re-verified	Metrics and confusion matrices for each condition
Model selection	Decision Tree for all three; hypertension to be re-confirmed	Weighted score 70% F1 + 30% accuracy
Streamlit dashboard	Completed	Condition selection, input collection and assessment results
Risk visualisation	Partially completed	Visualisations shown for the diabetes example; full set to be verified in the running application
External validation	Planned	Independent datasets and clinical validation not yet established
Deployment	Pending verification	Public deployment and production readiness not confirmed

13.3 Future Enhancements

- Re-run and document the hypertension pipeline from the source code, and consider a version of the task that does not use blood-pressure measurements as inputs.
- Add ROC-AUC, probability calibration, cross-validation and confidence intervals for all metrics.
- Add feature-importance or SHAP-based explanations for every module, including hypertension.
- Tune hyperparameters systematically and evaluate additional algorithms such as gradient boosting and support-vector machines.
- Validate the models on independent, representative datasets and assess subgroup performance and fairness.
- Document and test the heart-disease and hypertension dashboard flows, and add input validation and accessibility improvements.
- Support multi-condition risk summaries, so that a user can see the combined picture of diabetes, heart disease and hypertension indicators.

- Deploy with secure configuration, privacy safeguards, logging and clear clinical disclaimers, and involve clinicians in review.

13.4 Skills Demonstrated

The project demonstrates work with tabular healthcare data, preprocessing pipelines, supervised classification, evaluation metrics, model persistence, exploratory data analysis and Streamlit dashboard development, together with attention to the ethical communication of health information.

CHAPTER 14 – REFERENCES AND APPENDICES

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Appendix A – Feature Descriptions

A.1 Diabetes (Pima Indians Diabetes Database)

Feature	Description
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration from an oral glucose tolerance test
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin-fold thickness (mm)
Insulin	2-hour serum insulin (mu U/ml)
BMI	Body-mass index (weight in kg / height in m^2)
DiabetesPedigreeFunction	Score summarising diabetes family history
Age	Age in years
Outcome (target)	0 = no diabetes, 1 = diabetes

A.2 Heart disease (UCI Cleveland subset)

Feature	Description
age	Age in years
sex	Sex of the patient (coded)
cp	Chest-pain type
trestbps	Resting blood pressure (mm Hg)
chol	Serum cholesterol (mg/dl)
fbs	Fasting blood sugar above 120 mg/dl (coded)
restecg	Resting electrocardiographic result
thalach	Maximum heart rate achieved
exang	Exercise-induced angina
oldpeak	ST depression induced by exercise relative to rest
slope	Slope of the peak-exercise ST segment
ca	Number of major vessels coloured by fluoroscopy
thal	Thalassemia test result (coded)
target	Binary: 0 = absence, 1 = presence of heart disease

A.3 Hypertension (Framingham Heart Study dataset)

Feature	Description
male	Sex indicator (1 = male)
age	Age in years
currentSmoker	Whether the participant currently smokes
cigsPerDay	Average cigarettes smoked per day
BPMeds	Use of blood-pressure medication
prevalentStroke	History of stroke
diabetes	Diabetes indicator
totChol	Total cholesterol
sysBP	Systolic blood pressure
diaBP	Diastolic blood pressure
BMI	Body-mass index
heartRate	Heart rate
glucose	Glucose level
prevalentHyp (target)	0 = no prevalent hypertension, 1 = prevalent hypertension

Appendix B – Abbreviations

Abbreviation	Meaning
BMI	Body-mass index
CSV	Comma-separated values
EDA	Exploratory data analysis
FN / FP	False negative / false positive
TN / TP	True negative / true positive
ML	Machine learning
SHAP	Shapley additive explanations
UCI	University of California, Irvine
ROC-AUC	Area under the receiver-operating-characteristic curve

Appendix C – Reproducibility Settings

Setting	Value
Train–test split	80% / 20%, stratified
Random seed	random_state = 42 for the split and all three models
Imputation	SimpleImputer (median); zeros in Glucose, BloodPressure, SkinThickness, Insulin and BMI treated as missing for diabetes
Scaling	StandardScaler
Selection score	0.70 × F1-score + 0.30 × accuracy
Dashboard address (local)	localhost:8501

Appendix D – Source Code Repository

GitHub repository: https://github.com/Masuma-86/-DS_S7_T3_-HealthScope-A-Full-Lifecycle-Public-Health-Analytics-Risk-Dashboard